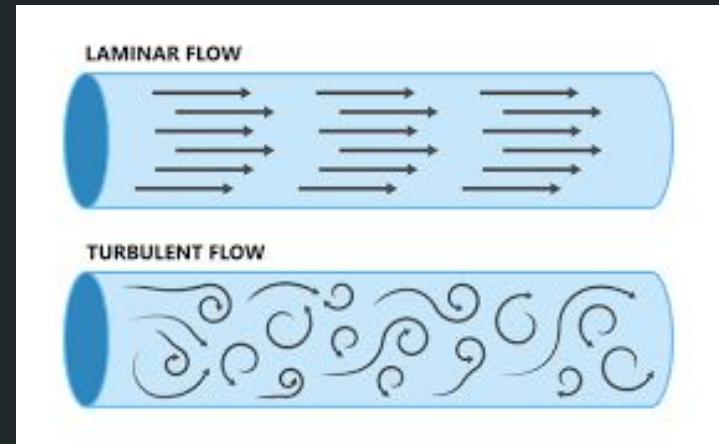


Using Chaos Theory To Understand the Motion of a Particle in Convection Currents

Jonathan Ramachandran

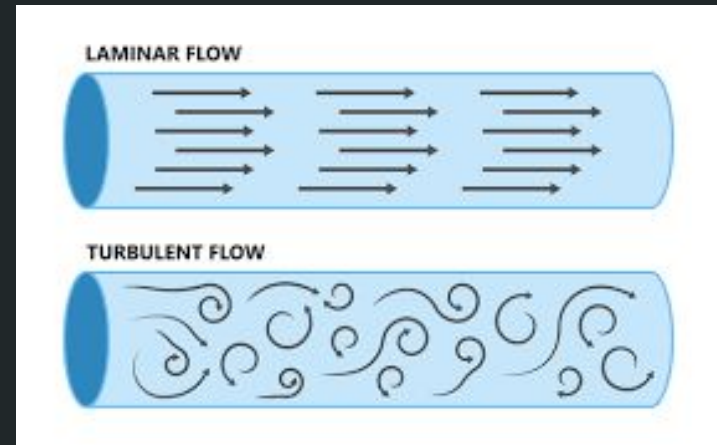
Introduction:

- “Turbulence is one of the most important unsolved problems in physics”
- Has vortexes, eddies, and frequent changes in velocity



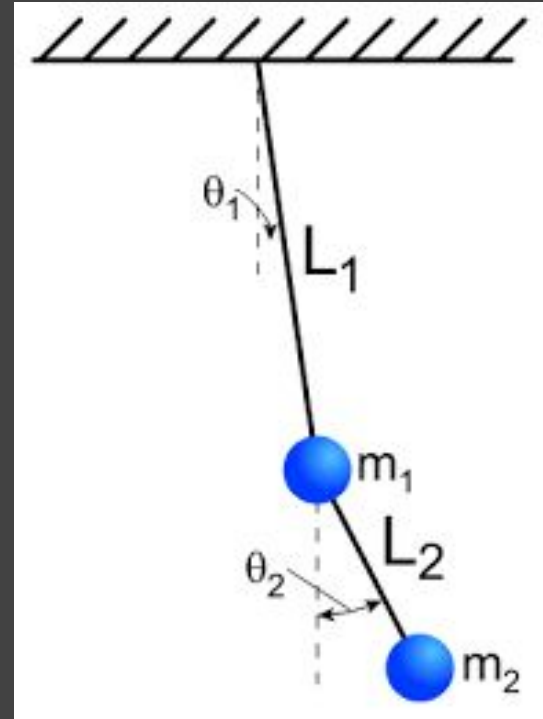
Solving Turbulence Would Mean:

- A drastic increase in energy efficiency for planes(neuralconcept.com)
- Billions of dollars saved by more efficient wind turbines(Discoverengineering.org)
- Better medicine delivery
- One million dollar prize and a Fields Medal for solving it



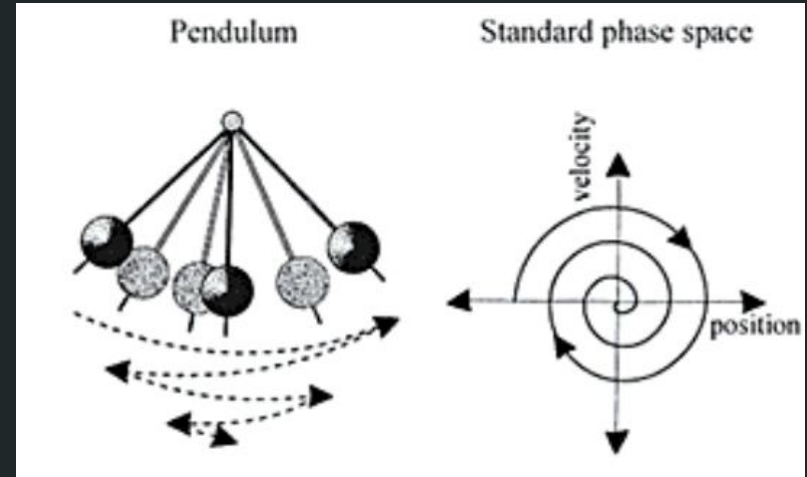
Chaos Theory:

- Chaotic System: A system that is fully deterministic, yet is hyper-sensitive to initial conditions
- Seemingly random, but contain order
- Turbulence is a classic example of a chaotic system
- Could predict turbulence if you knew everything about the system



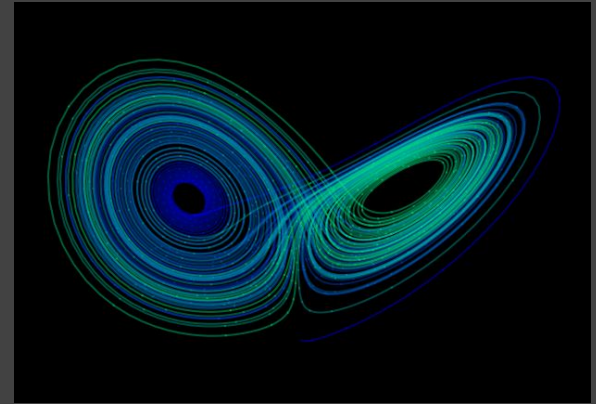
Phase Space:

- The set of all possible states of a system.
- A theoretical space
- Any point completely describes the system



Attractors:

- A set of states in phase space that the system tends to go towards
- Can be a point or a complex shape
- Chaotic systems produce strange attractors
- Shows order despite seeming randomness
- Reveal insights into seemingly random systems



The Lorenz Attractor

My Experiment:

- Making a turbulent environment through boiling water on the stove
- Releasing a particle in the middle and seeing if it makes a strange attractor in phase space (x, y, total velocity)
- Find order and patterns in the turbulent flow

Controlled variables:

- Location of particle release
- Temperature of water
- Centering pot on the stove
- The heat was on at the same temperature for all of them



Research Questions:

- **Will the motion of a particle moving in boiling water will form a strange attractor in phase space?(x,y, total velocity)**
- **Can chaos theory be applied to analyze the particle's attractor and reveal underlying patterns in its seemingly random behavior?**

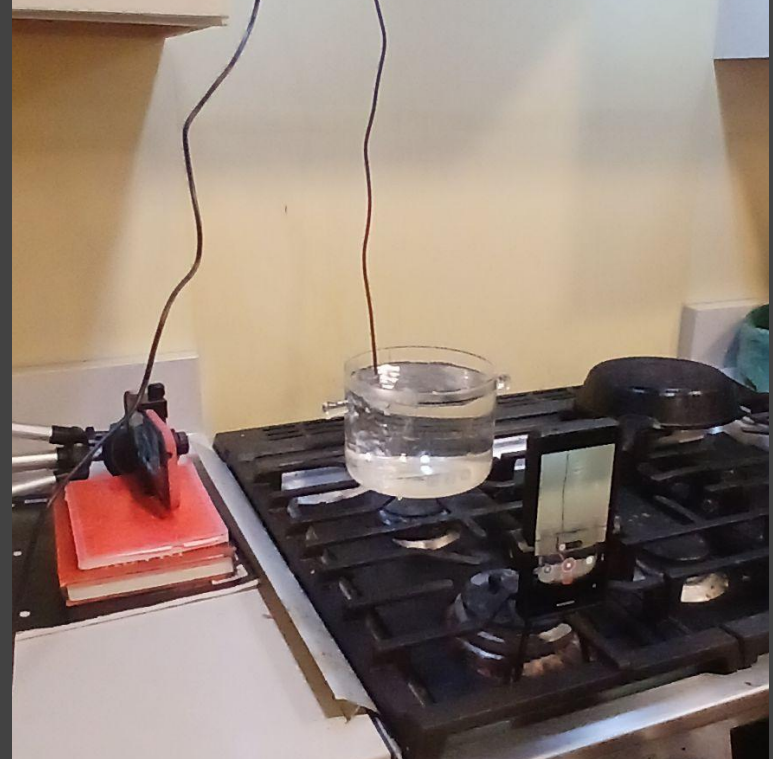
Materials

- Tongs
- A stove
- Water
- Thermometer
- Visual Studio Code
- Flashlight

- Two phone tripods
- Permanent Marker
- Glass 2 liter pot
- Tapioca pearls
- Red food coloring
- Two smartphones
- Sieve

Procedure- Experimentation

1. Use food coloring to dye tapioca seeds red.
2. Mark a cross on the back of the pot
3. Place the glass pot on the stove and turn the heat to medium.
4. Place a thermometer probe into the the water.
5. Start recording on both phones.
6. Shine a flashlight onto the water. This will help to synchronize the videos.



Procedure(continued)

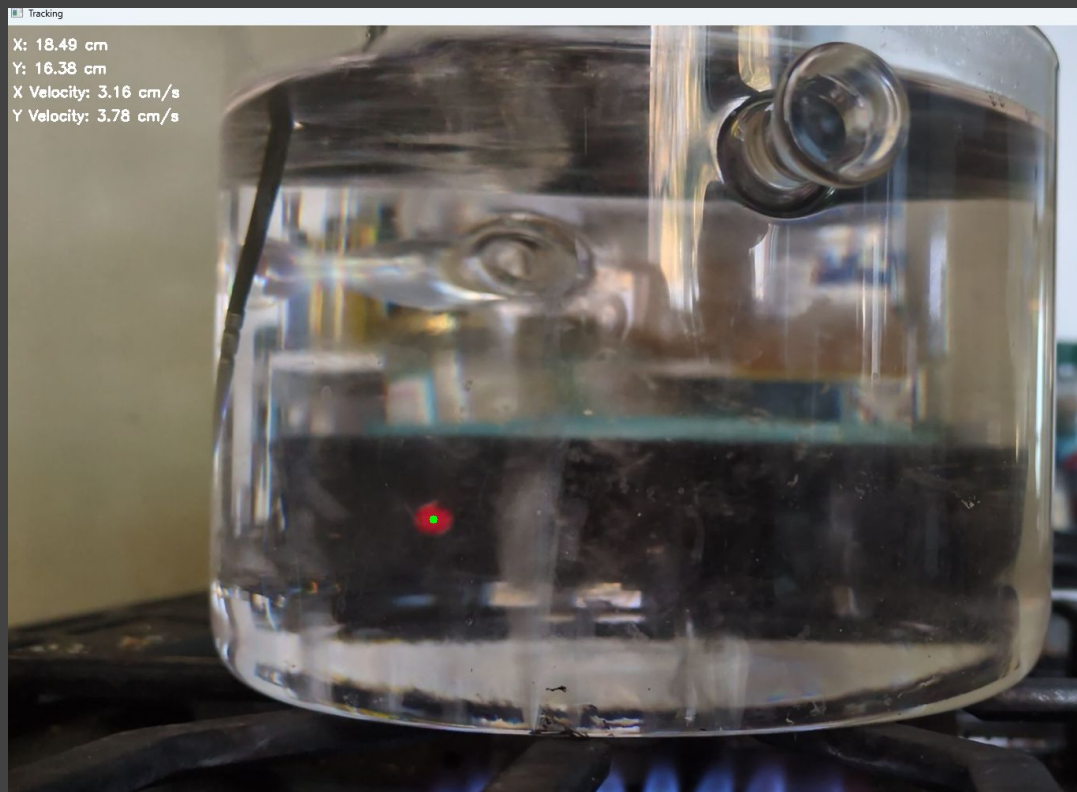
1. When the water reaches 206 degrees, use tongs to place the tapioca seed at the marked location and release.
2. Wait for two minutes.
3. Remove the particle with the sieve, stop the recording, and repeat thirty times.



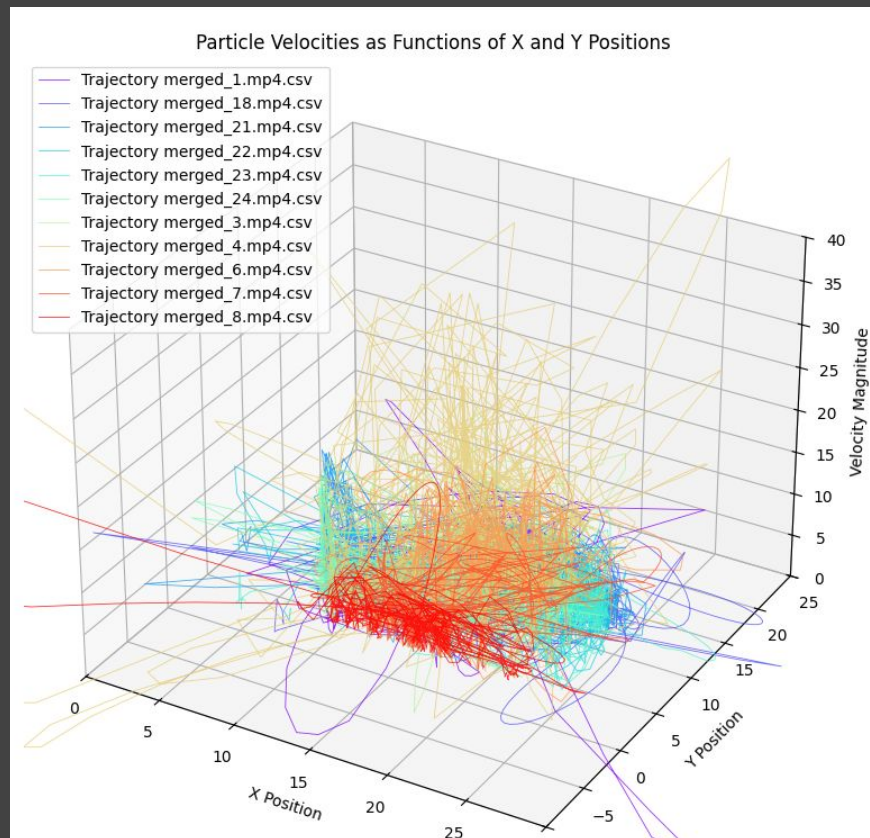
Procedure- Data Processing

1. Upload all the videos in order into a separate Google Drive folder for each one.
2. Find the starting time for each video by finding the timestamp when the particle is released into the water.
3. Generate a Python code that uses OpenCV to track the motion of a red particle.
4. Generate a Python code to clean the resulting data from outliers and nan values.
5. Generate a code that adds an x and y velocity column to each dataset from the side view, and adds a z velocity column to the front view.
6. Use a 'for' loop to loop to do the following for every video in first the front videos and then the side videos, passing the starting time of each video as an argument to the function.
7. Save all the CSVs to two separate folders.
8. Generate a script to combine the z velocity data column from the front datasets with the CSV datasets from the side videos. Save them in a new folder, arranged in numerical order.
9. Generate a Python script to loop through the folder and plot each one using Matplotlib.

OpenCV Tracking

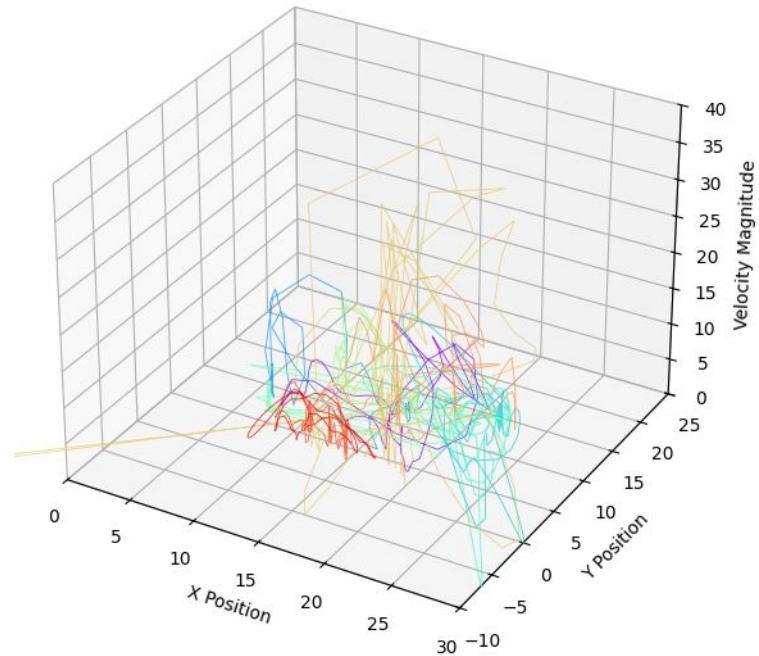


Results:



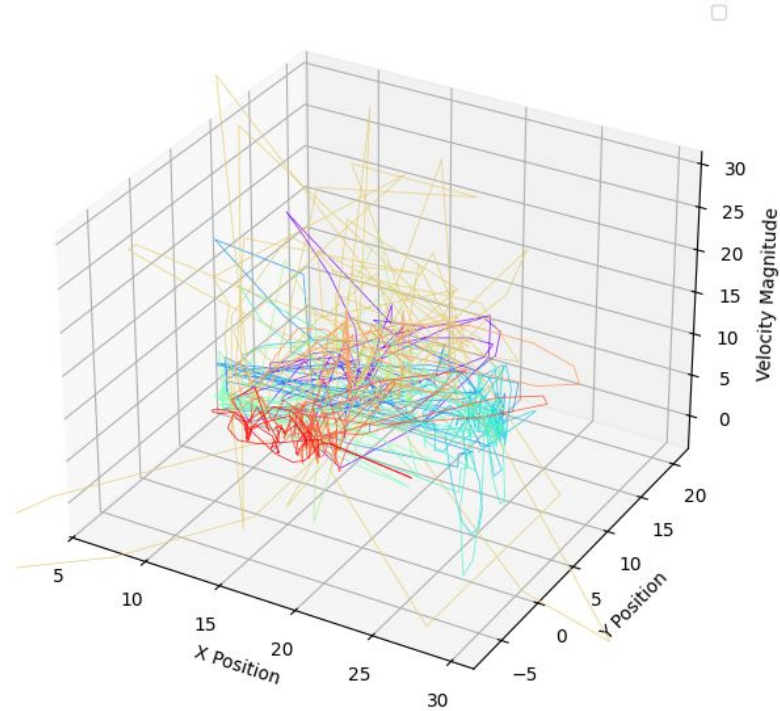
Results:

Particle Velocities as Functions of X and Y Positions (Indices 300-350)



Results:

Particle Velocities as Functions of X and Y Positions (Indices 200-300)



Statistical Analysis: Lyapunov Exponents

$$\delta(t) = \delta(0) \cdot e^{\lambda t}$$

Distance Between Trajectories 1 & 18 vs time:



Average Lyapunov Exponent across all data combinations (Slope of the fitted line):
0.000501 - System is Chaotic

No, a particle in convection currents will not make a strange attractor in phase space. What I would have done:

- Sets of states that the system tends towards
- Exact degree of chaos
- Time to chaos
- Zones of probability in the tank
- Analyze attractor geometry

Limitations and Areas of Further Research:

- Tank was too small
- Tapioca seeds sink
- OpenCV's failure to track points resulted in a loss of data
- Use Computational Fluid Dynamics to run simulations

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